

AUTOMOTIVE

Concentric connectors and Type 6 charging gun

Amphenol has introduced the DuraSwap concentric connectors and Type 6 charging gun solution, designed and



manufactured in India, to enhance the country's

e-mobility ecosystem. These solutions support two- and three-wheeler applications, ensuring safe, efficient charging for electric scooters and light vehicles are ideal for swappable battery systems, e-scooters, renewable energy, and energy storage devices.

Amphenol Communications Solutions
<https://www.amphenol-cs.com>

High-power MOSFETs

Renesas Electronics has launched 100V high-power N-Channel MOSFETs designed for high-current



switching applications such as motor control, battery management,

power management, and charging. These MOSFETs are ideal for electric vehicles, e-bikes, power tools, data

centres, and renewable energy systems like PV inverters. The MOSFETs support miniaturisation and flexible voltage systems for evolving electric vehicle and power conversion applications.

Renesas Electronics Corporation
<https://www.renesas.com/en>

Diodes for EV battery protection

Littelfuse's TPSMB-L series automotive TVS diodes are designed for



800V battery management systems (BMS) in electric vehicles (EVs). These diodes offer ultra-low clamping voltage, protecting

sensitive components like AFEs and BMICs in 14- to 20-cell BMS designs. Ideal for high-voltage EV applications, they ensure automotive-grade reliability, fast response times, and ISO-26262 compliance, enhancing the safety, performance, and longevity of BMS in advanced EV architectures.

Littelfuse, Inc.
<https://www.littelfuse.com>

semiconductor laser with a 420nm wavelength, promoting miniaturisation and cost reduction in optical systems.



Its high efficiency and long-term reliability are ideal for applications such as resin curing,

PCB manufacturing, 3D printing, and optical communication, providing eco-friendly, compact, and cost-effective solutions across various industries.

Nuvoton Technology Corporation
<https://www.nuvoton.com>

SoC for AI-powered computers

Qualcomm's Snapdragon X SoC targets AI-powered laptops and mini desktops, offering high performance at an

affordable price starting at \$600. The chip outperforms Intel's Core i5-1280U, supporting AI and machine learning tasks for Windows 11 devices from brands like



Acer, Dell, HP, and Lenovo, making AI-driven computing more accessible.
Qualcomm Technologies
<https://www.qualcomm.com>

Ultra processors

Intel has launched the Intel Core Ultra (Series 2) processors, enhancing mobile computing for businesses, creators, and gamers with AI improvements, efficiency, and performance. The Core Ultra 200V series, paired with Intel vPro, targets commercial

customers by offering advanced security, manageability, and productivity.



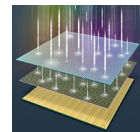
Intel also introduced the Core Ultra 200HX and H series for

creators and gamers, featuring up to 24 cores and Intel Arc graphics. Additionally, the Core Ultra 200U and 200S series focus on power efficiency and affordability.

Intel
<https://www.intel.com>

SPAD image sensors

Singular Photonics has launched advanced SPAD-based image sensors, integrating computation for low-light,



high-resolution imaging. Their Andarta sensor, developed with Meta, is ideal for medical imaging, enabling cerebral blood

flow monitoring in wearables. These sensors offer real-time processing, depth capture, and statistical analysis, transforming fields like healthcare, automotive, and science.

Singular Photonics
<https://singularphotonics.com>

Shunt resistors

Shunt resistors are crucial for efficient current sensing, balancing low resistance with high power handling. Ideal



for applications like electric vehicles, power tools, HVAC controls, and power supplies, the AEC-Q200-compliant

CSNL2512 ensures stable, accurate current measurement and efficient power management. High power density and reliability make it a cost-effective solution for demanding systems.

Stackpole
<https://www.seielect.com>

SMD ring core double chokes

TDK Corporation's EPCOS SurfIND series features current-compensated ring core double chokes designed for high currents and surface mounting.